

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented scalable data pipelines using **Python** and **SQL** to process onchain and offchain data for economic modeling, handling ~2,500 events per second.
- Developed and maintained production-grade economics applications and simulations, translating complex economic models into operational codebases.
- Collaborated with **Economics**, **Product**, and **Data Science** teams to gather requirements, understand network dynamics, and operationalize new incentive mechanisms.
- Engineered **Python** scripts for data extraction, transformation, and loading (ETL) from diverse sources, supporting automated reporting for economic analysis.
- Wrote and optimized complex **SQL** queries against large datasets for data validation, analysis, and the creation of curated datasets for economic simulations.
- Containerized data pipeline services using **Docker** and deployed them to a **Kubernetes** cluster on **AWS**, ensuring consistent and reproducible environments.
- Contributed to the design and parameterization of economics systems, analyzing simulation results to optimize network incentives and application performance.

SKILLS

Languages – Python, SQL, Java, R

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

Data Engineering & Orchestration – ETL, Data Pipelines, Airflow, DBT, Data Warehousing, Data Modeling

Databases – PostgreSQL, MySQL, Oracle

Software Engineering – Git, Docker, Kubernetes, CI/CD, Jenkins, REST APIs, Spring Boot

Analytics & ML – Pandas, NumPy, Scikit-learn, Power BI, Tableau, DAX, Statistical Analysis

PROJECTS

Onchain Data Pipeline for Economic Simulation

- Developed a **Python**-based data pipeline on **AWS Lambda** and **S3** to ingest and process real-time blockchain transaction data, serving as input for economic simulations.
- Implemented robust **SQL** transformations to aggregate and clean raw onchain data, creating feature sets that reduced data preparation time by **40%** for the economics team.
- Integrated the pipeline with **Airflow** for scheduling and monitoring, ensuring reliable data delivery for critical economic analysis and parameter tuning.

Incentive Mechanism Simulation Framework

- Designed and built a modular **Python** simulation framework to evaluate novel economic incentive schemes and network designs, capable of modeling up to 10,000 agents.
- Applied **Scikit-learn** to perform sensitivity analysis on simulation parameters, identifying key drivers of network participation and their impact on token velocity.
- Documented the framework's architecture and usage, enabling the economics team to independently run and analyze custom simulation scenarios.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)